Youkai Zhou

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EDUCATION

Drexel University Philadelphia, PA | Expected June 2028
Master of Science in Computer Science

University of Mississippi Oxford, MS | May 2026
Bachelor of Science in Computer Science | GPA: 3.5/4.0

TECHNICAL SKILLS

Programming and Web Languages: Python, SQL, JavaScript, TypeScript, Java, C++, Go, HTML
SQL Databases: PostgreSQL, SQLite, MySQL, DuckDB | NoSQL: Redis, MongoDB, DynamoDB
Machine Learning and Analytics: PyTorch, scikit-learn, pandas, NumPy, LightGBM, backtesting
Data and Application Tools: React, FastAPI, Kafka, Parquet, Fastify

TECHNICAL PROJECTS

Options Heatmap Oxford, MS | December 2024 - Present
Independent Developer | Personal project integrated into Prediction Stock

- Built React/TypeScript heatmaps from option-chain data, enabling comparisons across strikes and expirations
- Calculated gamma exposure from gamma and open interest, producing strike-level GEX summaries for options research
- Estimated VEX using provider vanna or Black-Scholes calculations, exposing vanna sensitivity alongside GEX analytics

Prediction Stock Oxford, MS | August 2024 - Present
Independent Developer | Integrated personal research platform

- Built immutable Parquet snapshots and DuckDB workflows, preserving point-in-time inputs for reproducible stock research
- Implemented LightGBM/PyTorch pipelines with purged walk-forward validation, enabling comparisons across time windows
- Connected FastAPI services to React/TypeScript views, unifying stock data, model evidence, and research diagnostics

Quant Model Oxford, MS | June 2022 - Present
Independent Developer | Personal project integrated into Prediction Stock

- Ingested crypto, U.S. equity, and China A-share data with Python, creating a shared workflow for multi-market research
- Implemented signal generation and backtesting for market research, enabling evaluation of historical trading signals
- Built Fastify APIs and React dashboards for quant research, making signals and backtest results accessible for review

WORK EXPERIENCE

Ichiban Asian Bistro and Go Madison, MS | May 2022 - August 2026
Seasonal employment: May-August and December-January

Assistant Manager December 2024 - August 2026

- Recorded kitchen and bar stock during seasonal shifts, sending inventory reports to the manager for review
- Checked kitchen equipment and front-desk computers during daily rounds, escalating faults for follow-up
- Coordinated staff schedules and opening/closing preparations for seasonal shifts, arranging coverage and order handoffs

Host and Front Desk Cashier May 2022 - December 2024

- Processed cash and card payments at the front desk during seasonal shifts, completing customer checkouts
- Coordinated reservations and seating for 150+ daily customers during seasonal service, assigning tables across the dining area
- Built an Excel tracker for seasonal front-desk reservations, consolidating booking records for seating management

SUPPORTING CONTENT

Relevant Coursework: Machine Learning, Data Mining, Database Systems, Data Structures and Algorithms, Statistics with Python, Software Engineering
Honor: Dean's List | Training: High Performance Computing Summer Institute, Jackson State University
Languages: English (fluent), Mandarin (native)